

粒子数可变系统

1. 只考虑一种粒子的情形 (单元系)

$$dU = TdS - pdV + u dN$$

u 化学势. 1 mol 的 G

$$u = \frac{G}{N} = u - Ts + pv$$

$$u = \frac{U}{N} \quad s = \frac{S}{N} \quad v = \frac{V}{N}$$

对于 1 mol 物质 (物质数量固定)

$$du = Tds - p dv$$

$$dU = d(Nu) = Ndu + u dN$$

$$= N(Tds - p dv) + u dN$$

$$\begin{cases} dS = d(Ns) = Nds + s dN \\ dV = d(Nv) = Ndv + v dN \end{cases}$$

$$\begin{cases} NTds = TdS - Ts dN \\ Np dv = p dV - pv dN \end{cases}$$

$$\begin{aligned} dU &= TdS - p dV + (u - Ts + pv) dN \\ &= TdS - p dV + u dN \end{aligned}$$

$$\text{当 } dN = 0 \text{ 时回到 } dU = Tds - PdV$$

$$2. \text{ 定义 } \begin{cases} H = U + PV \\ F = U - TS \\ G = U - TS + PV \end{cases}$$

基本微分方程为

$$\begin{cases} dU = Tds - PdV + u dN \\ dH = Tds + Vdp + u dN \\ dF = -SdT - PdV + u dN \\ dG = -SdT + Vdp + u dN \end{cases}$$

$$u = \left(\frac{\partial U}{\partial N} \right)_{S,V} = \left(\frac{\partial H}{\partial N} \right)_{S,p} = \left(\frac{\partial F}{\partial N} \right)_{T,V} = \left(\frac{\partial G}{\partial N} \right)_{T,p}$$

$$\text{由 } \begin{cases} du = Tds - pdv \\ Tds = d(Ts) - sdT \\ pdv = d(pv) - vdp \end{cases}$$

$$\Rightarrow du = d(Ts) - sdT - d(pv) + vdp$$

$$d(u - Ts + pv) = -sdT + vdp$$

$$du = -sdT + vdp$$

吉布斯—杜安关系

$$3. \text{ 由 } dF = -SdT - pdV + udn$$

$$(T, V, N) \rightarrow (T, V, u)$$

$$dF = -SdT - pdV + d(uN) - Nd u$$

$$d(F - uN) = -SdT - pdV - Nd u$$

$$\text{巨势 } \Psi = F - uN = F - G$$

$$d\Psi = -SdT - pdV - Nd u$$

热动平衡条件

1. 热动平衡判据：判断系统是否达到平衡态

热动平衡条件：维持热力学系统平衡态的具体条件。

① 热平衡条件：物体内部各部分不发生热量交换的条件

② 力学平衡条件：物体内部各部分之间不发生宏观位移的条件

③ 相变平衡条件：各相之间不发生物质转移(不发生相变)的条件。

④ 化学平衡条件: 化学反应不再进行的条件

2. 用熵判据推导平衡条件.

$$\begin{cases} \delta S = 0 \end{cases}$$

①

$$\delta^2 S < 0$$

② 稳定平衡, 充分

$$\delta U = 0, \delta V = 0, \delta N = 0$$

③

求平衡条件(平衡的必要条件), 只要用 ①. ③

假设系统由两个均匀子系统(相)组成. 两相之间可发生能量与物质交换, 每个相体积可变, 但总体积不变.

$$\begin{array}{cccc} S_1, S_2 & U_1, U_2 & V_1, V_2 & N_1, N_2 \\ \text{熵} & \text{内能} & \text{体积} & \text{摩尔数} \end{array}$$

对整个系统:

$$S = S_1 + S_2$$

$$U = U_1 + U_2$$

$$V = V_1 + V_2$$

$$N = N_1 + N_2$$

$$\delta S = \delta S_1 + \delta S_2 = \sum_{\alpha=1,2} \delta S_{\alpha}$$

$$S_{\alpha} = S_{\alpha}(U_{\alpha}, V_{\alpha}, N_{\alpha})$$

$$\delta S_{\alpha} = \left(\frac{\partial S_{\alpha}}{\partial U_{\alpha}} \right)_{V_{\alpha}, N_{\alpha}}^0 \delta U_{\alpha} + \left(\frac{\partial S_{\alpha}}{\partial V_{\alpha}} \right)_{U_{\alpha}, N_{\alpha}}^0 \delta V_{\alpha} + \left(\frac{\partial S_{\alpha}}{\partial N_{\alpha}} \right)_{U_{\alpha}, V_{\alpha}}^0 \delta N_{\alpha}$$

$$\textcircled{H} \quad dU_2 = T_2 dS_2 - P_2 dV_2 + \mu_2 dN_2$$

$$\Rightarrow dS_2 = \frac{1}{T_2} dU_2 + \frac{P_2}{T_2} dV_2 - \frac{\mu_2}{T_2} dN_2$$

$$\Rightarrow \left(\frac{\partial S_2}{\partial U_2} \right)_{V_2, N_2} = \frac{1}{T_2} \quad \left(\frac{\partial S_2}{\partial V_2} \right)_{U_2, N_2} = \frac{P_2}{T_2}$$

$$\left(\frac{\partial S_2}{\partial N_2} \right)_{U_2, V_2} = -\frac{\mu_2}{T_2}$$

$$\Rightarrow \delta S_2 = \frac{1}{T_2} \delta U_2 + \frac{P_2}{T_2} \delta V_2 - \frac{\mu_2}{T_2} \delta N_2$$

— 证明 δS_2 与 基本微分方程形式上相同。

$$\textcircled{H} \quad \delta U = 0 \Rightarrow \delta U_2 = -\delta U_1$$

$$\delta V = 0 \Rightarrow \delta V_2 = -\delta V_1$$

$$\delta N = 0 \Rightarrow \delta N_2 = -\delta N_1$$

$$\delta S = \delta S_1 + \delta S_2 = 0$$

$$= \left(\frac{1}{T_1} \delta U_1 + \frac{P_1}{T_1} \delta V_1 - \frac{\mu_1}{T_1} \delta N_1 \right) +$$

$$\left(\frac{1}{T_2} \delta U_2 + \frac{P_2}{T_2} \delta V_2 - \frac{\mu_2}{T_2} \delta N_2 \right)$$

$$= \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta U_1 + \left(\frac{P_1}{T_1} - \frac{P_2}{T_2} \right) \delta V_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2} \right) \delta N_1$$

$$\Rightarrow T_1 = T_2, \quad P_1 = P_2, \quad \mu_1 = \mu_2$$

3. 粒子数不守恒系统。

不存在约束 $N_1 + N_2 = N = \text{常数}$ ($\delta N \neq 0$)

$\delta N_2 \neq -\delta N_1$, δN_1 与 δN_2 可独立变化。

$$\begin{aligned} \text{由 } \delta S = & \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta U_1 + \left(\frac{p_1}{T_1} - \frac{p_2}{T_2} \right) \delta V_1 \\ & - \frac{\mu_1}{T_1} \delta N_1 - \frac{\mu_2}{T_2} \delta N_2 = 0 \end{aligned}$$

$$\Rightarrow T_1 = T_2, p_1 = p_2, \mu_1 = 0, \mu_2 = 0$$